

Tribhuvan University
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Cyber Law and Professional Ethics
BCA — Semester 7 — 2024 — Model Answer Sheet
Subject Code: CACS401

Note: This is a model answer sheet for study purposes. Answers are based on standard curriculum topics.

Group B

Attempt any SIX questions. [6×5=30]

2. Define cyber law. Explain the importance of cyber law in the digital age.

Answer: **Cyber Law** is the area of law dealing with the Internet, cyberspace, and computer systems. It covers intellectual property, data protection, privacy, and cybercrimes. **Importance:** 1) Protects individuals and organizations from online fraud and harassment. 2) Enables secure e-commerce and digital transactions. 3) Defines rights and responsibilities of internet users. 4) Provides legal framework for data protection. 5) Facilitates international cooperation on cybercrime. 6) Supports digital governance and e-government initiatives. Without cyber law, the digital economy would lack trust and security.

3. What is intellectual property? Explain the types of intellectual property rights.

Answer: **Intellectual Property (IP)** refers to creations of the mind protected by law. **Types:** **Copyright:** Protects literary, artistic, software works. Automatic, life+50-70 years. **Patent:** Protects inventions. Requires registration, 20 years. **Trademark:** Protects brand names, logos, slogans. Renewable indefinitely. **Trade Secret:** Protects confidential business information (formulas, processes). No expiration if kept secret. **Industrial Design:** Protects ornamental aspects of products.

4. Differentiate between copyright, patent, and trademark with examples.

Answer: **Copyright:** Protects expression of ideas (books, music, software). Automatic, no registration needed. Example: Windows source code. **Patent:** Protects inventions and processes. Must be novel, non-obvious, useful. Requires application and examination. Example: Google's search algorithm (if patented). **Trademark:** Protects brand identity. Prevents consumer confusion. Example: Apple logo, 'Nike' name. **Key Differences:** Copyright protects expression; patent protects invention; trademark protects brand. Copyright is automatic; patent requires registration; trademark benefits from registration.

5. Explain the Electronic Transaction Act 2063 of Nepal. What are its key provisions?

Answer: **Electronic Transaction Act 2063 (2008)** is Nepal's primary law governing electronic transactions and cybercrime. **Key Provisions:** 1) **Legal recognition** of electronic records and digital signatures. 2) **Controller** appointed to oversee implementation. 3) **License requirement** for certifying authorities issuing digital signatures. 4) **Cybercrime definitions:** Hacking, unauthorized access, damage to computer systems. 5) **Penalties:** Fines and imprisonment for cyber offenses. 6) **Evidence:** Electronic records admissible as evidence. 7) **Limitations:** Critics argue penalties are outdated and don't cover modern threats like ransomware, data

breaches.

6. What is cybercrime? Describe different types of cybercrimes with examples.

Answer: Cybercrime: Criminal activities conducted using computers or the Internet. **Types: Hacking:** Unauthorized access to systems. Example: Stealing data from a bank. **Phishing:** Fraudulent emails/websites to steal credentials. Example: Fake bank login pages. **Malware:** Viruses, ransomware, spyware. Example: WannaCry ransomware. **Identity Theft:** Stealing personal information for fraud. **Cyberstalking:** Online harassment. **Denial of Service:** Overwhelming servers to make services unavailable. **Child Exploitation:** Online abuse and trafficking. **Corporate Espionage:** Stealing trade secrets.

7. Define computer ethics. Explain the ten commandments of computer ethics.

Answer: Computer Ethics: Moral principles guiding computer use and professional conduct. **Ten Commandments (by Computer Ethics Institute):** 1) Thou shalt not use a computer to harm others. 2) Thou shalt not interfere with others' computer work. 3) Thou shalt not snoop around in others' files. 4) Thou shalt not use a computer to steal. 5) Thou shalt not use a computer to bear false witness. 6) Thou shalt not copy or use proprietary software without paying. 7) Thou shalt not use others' computer resources without authorization. 8) Thou shalt not appropriate others' intellectual output. 9) Thou shalt think about social consequences of programs. 10) Thou shalt use computers in ways that show consideration and respect.

8. What is a digital signature? How does it ensure authenticity and non-repudiation?

Answer: A Digital Signature is a cryptographic technique that verifies the authenticity and integrity of digital messages or documents. **How it works:** 1) Sender hashes the message. 2) Sender encrypts hash with their private key (signature). 3) Receiver decrypts signature with sender's public key. 4) Receiver hashes received message and compares. If match, message is authentic and unaltered. **Authenticity:** Only sender has private key. **Integrity:** Any change invalidates signature. **Non-repudiation:** Sender cannot deny signing because only they possess the private key. **Legal status:** Recognized by Electronic Transaction Act in Nepal and similar laws globally.

Group C

Attempt any TWO questions. [2×10=20]

9. Explain the concept of privacy in the digital age. Discuss the challenges to privacy posed by social media, surveillance, and data collection. How can individuals protect their privacy online?

Answer: Privacy in Digital Age: Right to control personal information and be free from unwarranted intrusion. **Challenges: Social Media:** Users share vast personal data voluntarily; platforms track behavior, build profiles, sell to advertisers. **Surveillance:** Governments and corporations monitor communications (CCTV, internet traffic, phone records). **Data Collection:** Cookies, trackers, IoT devices collect data 24/7. Big data analytics can infer sensitive information. **Protection Measures:** Use strong passwords and 2FA. Enable privacy settings on social media. Use VPNs and encrypted messaging (Signal). Review app permissions. Minimize data sharing. Use privacy-focused browsers (Firefox, Brave). Regularly audit online presence. Support privacy legislation (GDPR, data protection laws).

10. What is software licensing? Explain different types of software licenses (proprietary, open source, freeware, shareware). Discuss the GPL and MIT licenses.

Answer: Software Licensing: Legal agreement defining how software can be used, distributed, and modified. **Types: Proprietary:** Closed source, restrictions on use/modification. Example: Microsoft Windows. **Open Source:** Source code available, can modify/redistribute. **Freeware:** Free to use but source not available. Example: Adobe Reader. **Shareware:** Try before buy. Example: WinRAR. **GPL (GNU General Public**

License): Copyleft license. Derivative works must also be open source. Strong freedom protection. Used by Linux. **MIT License:** Permissive. Allows proprietary use with attribution. Minimal restrictions. Used by React, Node.js. **Comparison:** GPL ensures software freedom; MIT maximizes adoption and flexibility.

11. Describe the role of international organizations in governing cyberspace. Discuss WIPO, ITU, and their contributions to cyber law and internet governance.

Answer: International Organizations in Cyberspace Governance: **WIPO (World Intellectual Property Organization):** Administers international IP treaties. Manages domain name disputes (UDRP). Promotes copyright protection in digital environment. **ITU (International Telecommunication Union):** UN agency for ICT. Allocates global radio spectrum and satellite orbits. Develops technical standards. Promotes broadband access and cybersecurity. **ICANN:** Manages domain name system and IP address allocation. **ISO/IEC:** Develops security standards (ISO 27001). **Challenges:** Sovereignty conflicts, varying national laws, enforcement difficulties, balancing security with freedom. Need for multistakeholder approach involving governments, private sector, civil society.

These model answers are prepared for educational purposes. Refer to textbooks for comprehensive study.